

# Erik Jonsson School of Engineering and Computer Science

## *Department of Computer Science*

### Computer Science (BS)

The Computer Science Department offers the [BS degree in Computer Science](#) and the [BS degree in Software Engineering](#). Both are based on a solid foundation of mathematics, including calculus, and discrete mathematics. These programs of study are designed to offer students opportunities to prepare for an industrial, business, or governmental career in a rapidly changing profession and to prepare for graduate study in a field in which further education is strongly recommended. The two programs have the same basis in core computer science, including the analysis of algorithms and data structures, modern programming methodologies, and the study of operating systems. The Computer Science program continues with courses in advanced data structures, programming languages, telecommunications networks, and automata theory, while the Software Engineering program includes courses in requirements engineering, software validation and testing, and software architecture, culminating in a challenging project course in which students must demonstrate use of software engineering techniques. Both programs offer a rich choice of elective studies, including courses in artificial intelligence, computer graphics, and compiler design.

The school offers a "fast track" BS / MS option; see [Fast Track Baccalaureate/Master's Degree Program](#).

The Computer Science Department and the Department of Mathematical Sciences in the School of Natural Sciences and Mathematics jointly offer a [BS degree in Data Science](#).

### Mission of the Department of Computer Science

The mission of the Department of Computer Science is to prepare undergraduate and graduate students for productive careers in industry, academia, and government by providing an outstanding environment for teaching, learning, and research in the theory and applications of computing. The Department places a high priority on establishing and maintaining innovative research programs to enhance its education quality and make it an important regional, national, and international resource center for discovering, integrating, and applying new knowledge and technologies.

## Goals for the Computer Science Program

The undergraduate Computer Science program is committed to providing students with a high-quality education and preparing them for long and successful careers in industry and government.

Our graduates, while eminently ready for immediate employment, will also be fully ready for focused training as required for specific positions in Computer Science and closely related areas. Graduates interested in highly technical careers, research, and/or academia will be fully prepared to further their education in graduate school.

## Program Educational Objectives for Computer Science

Within a few years after graduation, graduates of the Computer Science program should:

- Pursue lasting and successful careers in computer science or related fields including as industry professionals, educators, researchers, government professionals and/or entrepreneurs
- Be effective team contributors and/or leaders, including in multi-disciplinary environments, in professional fields related to computer science
- Serve their profession in a socially responsible, ethical, and innovative manner to meet the software engineering needs of industry, academia, and/or society
- Continue to develop and grow through lifelong learning, and successfully adapt to continuous advancements in software technology

## ABET Accreditation

The BS program in Computer Science is accredited by the Computing Accreditation Commission of ABET, [www.abet.org](http://www.abet.org).

## Bachelor of Science in Computer Science

[Degree Requirements](#) (124 semester credit hours)<sup>1</sup>

[View an Example of Degree Requirements by Semester](#)

## Faculty

FACG> ecs-computer-science-bs

**Professors:** Farokh B. Bastani, Sergey Bereg, Ovidiu Daescu, Yvo G. Desmedt, Ding-Zhu Du, András Faragó, Paul Fishwick, Xiaohu Guo, Gopal Gupta, Zygmunt Haas, Kevin Hamlen, Sanda M. Harabagiu, Dung T. Huynh, Jason Jue, Murat Kantarcioglu, Latifur Khan, Andrian Marcus, Neeraj Mittal, Dan I. Moldovan, Sriraam Natarajan, Yu-Chung (Vincent) Ng, Tien Nguyen, Simeon C. Ntafos,

Balakrishnan Prabhakaran, Ravi Prakash, Kamil Sarac, Haim Schweitzer, Bhavani Thuraisingham, W. Eric Wong, Weili Wu, I-Ling Yen, Rym Zalila-Wenkstern

**Associate Professors:** Feng Chen, Lawrence Chung, Jorge A. Cobb, Vibhav Gogate, Benjamin Raichel, Nicholas Ruozzi

**Assistant Professors:** Xinya Du, Emily Kyle Fox, Yunhui Guo, Shuang Hao, Rishabh Iyer, Kangkook Jee, Chung Hwan Kim, Jin Kim, Jessica Ouyang, Yapeng Tian, Shiyi Wei, Yu Xiang, Wei Yang

**Professors Emeriti:** R. Chandrasekaran, Ivor P. Page, William J. Pervin, Balaji Raghavachari, Ivan Hal Sudborough, Klaus Truemper, Subbarayan Venkatesan, Kang Zhang

**Professors of Instruction:** Ebru Cankaya, John Cole, Doug DeGroot, Timothy (Tim) Farage, Shyam Karrah, Pushpa Kumar, Nhut Nguyen, Greg Ozbirn, Miguel Razo-Razo, Jeyakesavan (Jey) Veerasamy

**Associate Professors of Instruction:** Sridhar Alagar, Gordon Arnold, Anjum Chida, Wei Pang Chin, Bhadrachalam Chitturi, Michael Christiansen, Chris I. Davis, Karen Doore, Neeraj Gupta, Khiem Le, Anarag Nagar, Mehra Nouroz Borazjany, Jalal Omer, Mark Paulk, Jason W. Smith, Laurie Thompson, James Willson, Nurcan Yuruk

**Assistant Professors of Instruction:** Eric Becker, Scott Dollinger, Serdar Erbatur, Ranran Feng, Omar Hamdy, Gity Karami, Kamran Khan, Karen Mazidi, Richard K. Min, Priya Narayanasami, Brian Ricks, Elmer Salazar, Meghana Satpute, Nidhiben Solanki, Srimathi Srinivasan, Yi Zhao

I. Core Curriculum Requirements: 42 semester credit hours<sup>2</sup>

**Communication: 6 semester credit hours**

[RHET 1302](#) Rhetoric<sup>3</sup>

[ECS 3390](#) Professional and Technical Communication<sup>3</sup>

Or select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

**Mathematics: 3 semester credit hours**

[MATH 2413](#) Differential Calculus<sup>4</sup>

or [MATH 2417](#) Calculus I<sup>4</sup>

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

**Life and Physical Sciences: 6 semester credit hours**

[PHYS 2325](#) Mechanics<sup>5</sup>

[PHYS 2326](#) Electromagnetism and Waves<sup>5</sup>

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

**Language, Philosophy and Culture: 3 semester credit hours**

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

**Creative Arts: 3 semester credit hours**

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

**American History: 6 semester credit hours**

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

**Government/Political Science: 6 semester credit hours**

[GOVT 2305](#) American National Government

[GOVT 2306](#) State and Local Government

Or select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

**Social and Behavioral Sciences: 3 semester credit hours**

Select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

**Component Area Option: 6 semester credit hours**

[MATH 2413](#) Differential Calculus<sup>4</sup>

or [MATH 2417](#) Calculus I<sup>4</sup>

[MATH 2419](#) Calculus II<sup>4</sup>

[PHYS 2125](#) Physics Laboratory I<sup>5</sup>

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

**II. Major Requirements: 72 semester credit hours**

**Major Preparatory Courses: 24 semester credit hours beyond Core Curriculum**

[ECS 1100](#) Introduction to Engineering and Computer Science<sup>6</sup>

[CS 1200](#) Introduction to Computer Science and Software Engineering

[CS 1436](#) Programming Fundamentals

[CS 1337](#) Computer Science I

[CS 2305](#) Discrete Mathematics for Computing I<sup>7</sup>

[CS 2336](#) Computer Science II

[CS 2340](#) Computer Architecture

MATH Sequence - Students may choose one of the following sequences:

I. [MATH 2413](#) Differential Calculus<sup>4</sup>

and [MATH 2414](#) Integral Calculus<sup>4</sup>

or

II. [MATH 2417](#) Calculus I<sup>4</sup>

and [MATH 2419](#) Calculus II<sup>4</sup>

[MATH 2418](#) Linear Algebra

[RHET 1302](#) Rhetoric<sup>3</sup>

[PHYS 2125](#) Physics Laboratory I<sup>5</sup>

[PHYS 2126](#) Physics Laboratory II

[PHYS 2325](#) Mechanics<sup>5</sup>

[PHYS 2326](#) Electromagnetism and Waves<sup>5</sup>

**Major Core Courses: 39 semester credit hours beyond Core Curriculum**

[CS 3162](#) Professional Responsibility in Computer Science and Software Engineering

[CS 3305](#) Discrete Mathematics for Computing II

[CS 3341](#) Probability and Statistics in Computer Science and Software Engineering<sup>8</sup>

[CS 3345](#) Data Structures and Introduction to Algorithmic Analysis

[CS 3354](#) Software Engineering

[CS 3377](#) Systems Programming in UNIX and Other Environments

[ECS 3390](#) Professional and Technical Communication<sup>3</sup>

[CS 4141](#) Digital Systems Laboratory

[CS 4337](#) Programming Language Paradigms

[CS 4341](#) Digital Logic and Computer Design

[CS 4347](#) Database Systems

[CS 4348](#) Operating Systems Concepts

[CS 4349](#) Advanced Algorithm Design and Analysis

[CS 4384](#) Automata Theory

[CS 4485](#) Computer Science Project

### **Major Guided Electives: 9 semester credit hours**

CS guided electives are 4000 level CS courses approved by the student's CS advisor. The following courses may be used as guided electives without the explicit approval of an advisor:

[CS 4314](#) Intelligent Systems Analysis

[CS 4315](#) Intelligent Systems Design

[CS 4334](#) Numerical Analysis

[CS 4336](#) Advanced Java

[CS 4352](#) Introduction to Human-Computer Interaction

[CS 4361](#) Computer Graphics

[CS 4365](#) Artificial Intelligence

[CS 4375](#) Introduction to Machine Learning

[CS 4376](#) Object-Oriented Design

[CS 4386](#) Compiler Design

[CS 4389](#) Data and Applications Security

[CS 4390](#) Computer Networks

[CS 4391](#) Introduction to Computer Vision

[CS 4392](#) Computer Animation

[CS 4393](#) Computer and Network Security

[CS 4394](#) Implementation of Modern Operating Systems

[CS 4395](#) Human Language Technologies

[CS 4396](#) Networking Laboratory

[CS 4397](#) Embedded Computer Systems

[CS 4398](#) Digital Forensics

[CS 4399](#) Senior Honors in Computer Science

[EE 4325](#) Introduction to VLSI Design

[SE 4351](#) Requirements Engineering

[SE 4352](#) Software Architecture and Design

[SE 4367](#) Software Testing, Verification, Validation and Quality Assurance

[SE 4381](#) Software Project Planning and Management

Any other organized CS 4300-level course

III. Elective Requirements: 10 semester credit hours

**Free Electives: 10 semester credit hours**

Both lower- and upper-division courses may count as free electives.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

## Fast Track Baccalaureate/Master's Degrees

In response to the need for post-baccalaureate education in the exciting field of computer science, a Fast Track program is available to well-qualified UT Dallas undergraduate students. Qualified seniors may take up to 15 graduate semester credit hours that may be used to complete the baccalaureate degree and also to satisfy requirements for the master's degree. Interested

students should see the Associate Dean of Undergraduate Education (ADU) for specific requirements.

## Honors Programs

The Department of Computer Science offers an Honors Program called Computer Science Computing Scholars (CS<sup>2</sup>). (CS<sup>2</sup>) is an intense Bachelor of Science in Computer Science Degree Program created for exceptionally gifted students who wish to pursue a demanding course of study enriched throughout with research experiences. The Computing Scholars Program has a specially designed curriculum. Courses integrate discussion of current research, recent discoveries, and open problems into a rich logical progression of firmly related topics. Course numbers for the Core Curriculum Requirements and Major Requirements are the same as those for the Bachelor of Science in Computer Science, but Computing Scholars take honors versions of the following courses: [CS 1200](#), [CS 2305](#), [CS 2340](#), [CS 3305](#), [CS 3341](#), [CS 3345](#), [CS 3354](#), [CS 4141](#), [CS 4337](#), [CS 4341](#), [CS 4348](#), [CS 4349](#), and [CS 4384](#).

Admission to the program is mainly by nomination and invitation. Those invited to join the Computing Scholars Honors Program will have successfully completed a full and challenging high school curriculum, will have achieved high scores on the SAT or ACT tests, and will be about to graduate from high school, or equivalent, with high class rank.

Successful participants will graduate with the added distinction of a Computing Scholars Honors Diploma.

For more information about this program students should contact the Computer Science Department leadership.

## Minors

Students will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average).

### Minor in Computer Science

21 semester credit hours

Students majoring in Computer Engineering, Computer Science or Software Engineering cannot add a minor in Computer Science.

A minor in Computer Science requires 21 semester credit hours earned through the following courses:

[CS 1337](#) Computer Science I



[CS 2305](#) Discrete Mathematics for Computing I

[CS 2336](#) Computer Science II

[CS 3305](#) Discrete Mathematics for Computing II

[CS 3345](#) Data Structures and Introduction to Algorithmic Analysis

[CS 3354](#) Software Engineering

CS Elective (any 4000-level organized CS class)

## Minor in Information Assurance

30 semester credit hours

A minor in Information Assurance requires 30 semester credit hours earned through the following courses:

[CS 1337](#) Computer Science I

[CS 2305](#) Discrete Mathematics for Computing I

[CS 2336](#) Computer Science II

[CS 3305](#) Discrete Mathematics for Computing II

[CS 3345](#) Data Structures and Introduction to Algorithmic Analysis

[CS 4347](#) Database Systems

[CS 4348](#) Operating Systems Concepts

[CS 4389](#) Data and Applications Security

[CS 4393](#) Computer and Network Security

[CS 4398](#) Digital Forensics

## Certificates

### Certificate in Information Assurance

9 semester credit hours

A Certificate in Information Assurance is offered by the Department of Computer Science. It can be obtained by completing the following (as well as any required prerequisites):

[CS 4389](#) Data and Applications Security

[CS 4393](#) Computer and Network Security

[CS 4398](#) Digital Forensics

Students that complete the Minor in Information Assurance will not be awarded certificates in Information Assurance.

1. Incoming freshmen must enroll and complete requirements of UNIV 1010 and the corresponding school-related freshman seminar course. Students, including transfer students, who complete their core curriculum at UT Dallas must take UNIV 2020.
2. Curriculum Requirements can be fulfilled by other approved courses. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.
3. Semester credit hours fulfill the communication component of the Core Curriculum.
4. Three semester credit hours of Calculus are counted under Mathematics Core, and five semester credit hours of Calculus are counted as Component Area Option Core.
5. Six semester credit hours of Physics are counted under Science core, and one semester credit hour of Physics (PHYS 2125) is counted as Component Area Option Core.
6. Transfer students with sufficient background may petition to substitute upper-division semester credit hours in the major for this class.
7. BS in Data Science students can substitute MATH 3315 for CS 2305.
8. BS in Data Science students can substitute STAT 3355 for CS 3341.

Updated: 2023-05-30 16:39:25 v15.a90a4b